

- 12 **(long)** (a) A soft iron bar with a square cross-section is placed touching the ends of a C-shaped soft iron bar with a square cross-section, as shown in the side elevation of Fig. 9, in which all dimensions are in mm. The relative permeability of both bars is fixed at 1000. A coil with 20 turns is wrapped around the straight bar. What is the approximate magnetic flux density at the centre of the straight bar if a current of 2 A is passed through the coil? [10]
- (b) If the weights of the straight bar and coil are negligible, what is the maximum weight which can be suspended from the straight bar (assuming that it is in contact with the C-shaped bar) without it being pulled from the C-shaped bar? [10]
- (c) The straight and C-shaped bar are now separated so that there is a gap of 2 mm between the two. What is the approximate magnetic flux density at the centre of the straight bar now? [10]

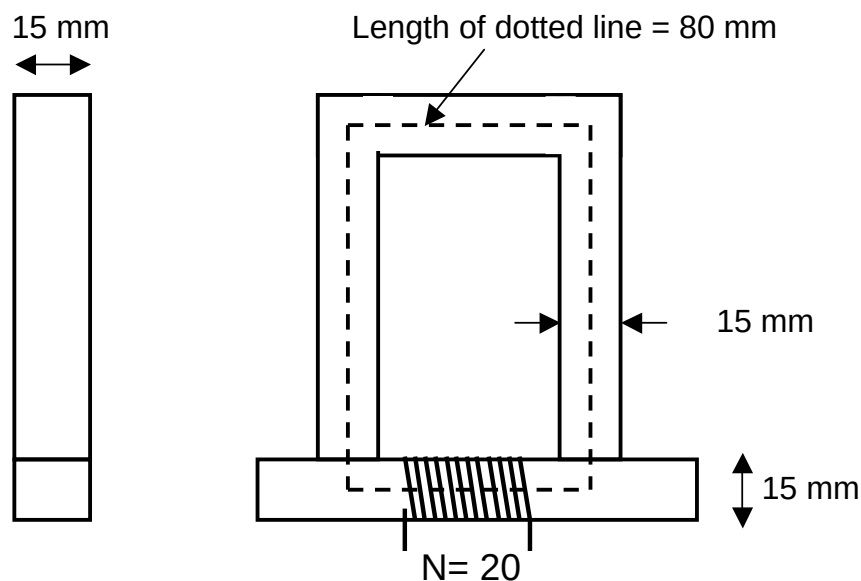


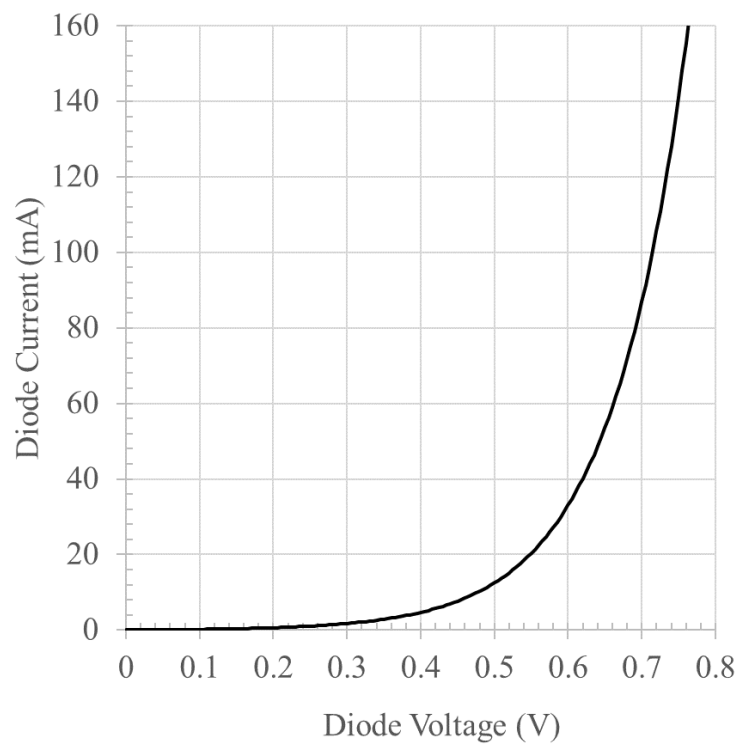
Fig. 9

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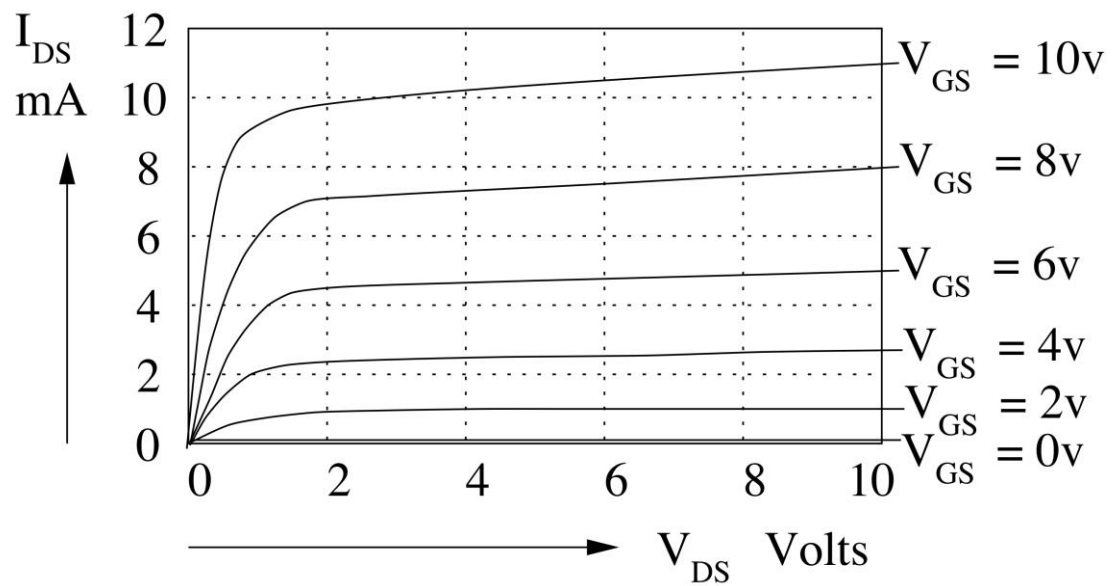


Extra copy of Fig. 2 for Question 1

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Extra copy of Fig. 7(a) for Question 6